

INTRODUCCION A LA CRIPTOGRAFIA MODERNA

Homework 1

Depto. de Computación, FCEN, UBA, 1er. Bimestre 2026

26/3/26

Student:

Due: 7/4/25, 23:59 hs

Instructions

- Upload your solution to Campus; make sure it's only one file, and clearly write your name on the first page. Name the file '<your last name>_HW1.pdf.'
If you are proficient with $\text{\LaTeX}$, you may also typeset your submission and submit in PDF format. To do so, uncomment the "`%\begin{solution}`" and "`%\end{solution}`" lines and write your solution between those two command lines.
- Your solutions will be graded on *correctness* and *clarity*. You should only submit work that you believe to be correct.
- You may collaborate with others on this problem set. However, you must **write up your own solutions** and **list your collaborators and any external sources (including ChatGPT and similar generative AI chatbots)** for each problem. Be ready to explain your solutions orally to a member of the course's staff if asked.

This assignment contains 6 questions, for a total of 60 points.

1. We saw the following definition of *negligible* functions in class:

Definition: A positive function $\mu(n)$ is called negligible if for any positive polynomial $p(n)$, there exists a constant n_0 such that for all $n \geq n_0$ we have $\mu(n) < 1/p(n)$.

- (a) (6 points) Determine whether each of the following functions is negligible. Justify your answers.

i. $f(n) = 2^{-(\log n)(\log \log n)}$

Solution: Para verificar que $f(n) = 2^{-(\log n)(\log \log n)}$ es negligible, debemos mostrar que es $O(1/p(n))$ para todo polinomio positivo $p(n)$. Veamoslo con el límite, si es que existe. Si el límite es ∞ , entonces pertenece a $O(1/p(n))$, por ende es negligible.

$$\lim_{n \rightarrow \infty} \frac{2^{-(\log n)(\log \log n)}}{1/p(n)} = \lim_{n \rightarrow \infty} \frac{p(n)}{2^{(\log n)(\log \log n)}} = \lim_{n \rightarrow \infty} \frac{p(n)}{n^{\log \log n}}$$

Vemos que, como $p(n)$ es un polinomio positivo, crece como n^k para algún $k > 0$. Como el denominador crece como $n^{\log \log n}$, el exponente no es fijo, y como $n \rightarrow \infty$, el exponente tiende a ∞ , mientras que k es fijo. Por ende, el límite es 0, por ende $f(n)$ es negligible.

ii. $f(n) = n^{-1/n}$

Solution: Hacemos lo mismo que en el inciso anterior.

$$\lim_{n \rightarrow \infty} \frac{n^{-1/n}}{1/p(n)} = \lim_{n \rightarrow \infty} \frac{p(n)}{n^{1/n}} = \infty$$

Vemos que en este caso, el denominador tiende a 0 y el numerador tiende a ∞ , luego el límite es ∞ y $f(n)$ no es negligible.

iii. $f(n) = 1/n$ for odd n and $f(n) = 2^{-n}$ for even n

Solution: En este caso, como la función es partida, tenemos que verificar que ambas son $O(1/p(n))$ para todo polinomio positivo $p(n)$. Veamos primero el caso n impar.

$$\lim_{n \rightarrow \infty} \frac{1/n}{1/p(n)} = \lim_{n \rightarrow \infty} \frac{p(n)}{n} = \infty$$

Luego, como tiende a ∞ , no es $O(1/p(n))$; luego no es negligible.

(b) (4 points) We saw that the definition can be restated as follows:

Lemma: A positive function $\mu(n)$ is negligible if and only if $\mu(n) \in O(1/p(n))$ for all positive polynomials p .

Prove the reverse ($\Leftarrow$) direction of this statement.

Solution: Quiero ver que vale la siguiente implicación:

$$\mu(n) \in O(1/p(n)) \text{ para todo } p(n) \text{ positivo} \implies \mu(n) \text{ es negligible}$$

La definición de negligible es que, para todo polinomio positivo $p(n)$, existe un n_0 tal que para todo $n \geq n_0$ se cumple que $\mu(n) < 1/p(n)$. Esto es muy similar a la definición de O , pero sin la constante $c > 0$.

$$\begin{aligned} \forall p(n) > 0, \exists n_0, c > 0, \forall n \geq n_0, \mu(n) < c \cdot 1/p(n) \\ \implies \forall q(n) > 0, \exists n_1, \forall n \geq n_1, \mu(n) < 1/q(n). \end{aligned}$$

Luego, como lo de la izquierda vale para cualquier polinomio, lo tomamos para un polinomio particular, con el objetivo de llegar a la expresión de la derecha. Tomamos $n \cdot q(n)$, para un polinomio $q(n)$ positivo cualquiera. Luego,

$$\mu(n) < \frac{c}{n} \cdot \frac{1}{q(n)}$$

Entonces, para llegar a lo de la derecha, queremos mostrar que

$$\frac{c}{n} \cdot \frac{1}{q(n)} < 1/q(n)$$

Y esto sucede cuando $\frac{c}{n} < 1$, o sea, cuando $c < n$. Luego, para llegar a lo de la derecha, podemos tomar $n_1 = \max(n_0, c)$. Entonces, para todo $n \geq n_1$, se cumple que $\mu(n) < 1/q(n)$, por ende $\mu(n)$ es negligible. $\square$

2. (10 points) Assume an attacker knows that a user's password is either abcd or bedg. Say the user encrypts his password using the shift cipher, and the attacker sees the resulting ciphertext. Show how the attacker can determine the user's password, or explain why this is not possible.
3. (10 points) We proved in class that if a cipher is *perfectly secure/secret*, then it is *indistinguishable*. Prove the **opposite** direction. Begin by stating the perfect security and indistinguishability properties.

Solution: Un cipher es perfecto si:

$$P(\text{plaintext} = m | \text{ciphertext} = c) = P(\text{plaintext} = m), \text{ para cualquier } m \in \mathcal{M}.$$

Un cipher es indistinguible si:

$$\forall m_1, m_2, c, P(\text{Enc}(m_1) = c) = P(\text{Enc}(m_2) = c)$$

Queremos ver que si es indistinguible, entonces es perfecto. Es decir:

$$\begin{aligned} \forall m_1, m_2 \in \mathcal{M}, \forall c : P(\text{Enc}_K(m_1) = c) &= P(\text{Enc}_K(m_2) = c) \\ \implies \forall m, c : P(M = m | C = c) &= P(M = m) \end{aligned}$$

Donde M es el plaintext, C es el ciphertext y $\mathcal{M}$ es el espacio de mensajes.

Sea $m \in \mathcal{M}$ y c un ciphertext con $P(C = c) > 0$. Por Bayes:

$$P(M = m | C = c) = \frac{P(C = c | M = m) \cdot P(M = m)}{P(C = c)}$$

Luego, basta ver que $P(C = c | M = m) = P(C = c)$, de ese modo se cancela con el denominador y queda $P(M = m | C = c) = P(M = m)$, que es lo que queremos demostrar.

Vemos que $P(C = c | M = m) = P(\text{Enc}(m) = c)$. Vemos que esta probabilidad es igual para todo $m \in \mathcal{M}$, por indistinguibilidad. Llamamos a este valor α , independientemente de qué m tomemos.

Veamos ahora que $P(C = c) = \alpha$.

Por probabilidad total:

$$P(C = c) = \sum_{m' \in \mathcal{M}} P(C = c \mid M = m') \cdot P(M = m')$$

Luego,

$$P(C = c) = \sum_{m' \in \mathcal{M}} P(\text{Enc}(m') = c) \cdot P(M = m')$$

Reemplazamos por α :

$$P(C = c) = \alpha \cdot \sum_{m' \in \mathcal{M}} P(M = m') = \alpha \cdot 1 = \alpha$$

Y también $P(C = c \mid M = m) = P(\text{Enc}(m) = c) = \alpha$.

Luego $P(C = c \mid M = m) = P(C = c)$, y sustituyendo en Bayes:

$$P(M = m \mid C = c) = \frac{\alpha \cdot P(M = m)}{\alpha} = P(M = m)$$

Por lo tanto, el cipher es perfectamente seguro. □

4. We look at the different conditions under which the shift and mono-alphabetic substitution ciphers are perfectly secret:
 - (a) (4 points) Prove that if only a single character is encrypted, then the shift cipher is perfectly secret.
 - (b) (3 points) Does the perfectly secret conditional probability equation hold for the mono-alphabetic substitution cipher? Elaborate.
 - (c) (3 points) What is the largest message space $\mathcal{M}$ for which the mono-alphabetic substitution cipher provides perfect secrecy?
5. (10 points) Suppose you have a message m of length n , but you can only generate random keys of length k and l such that $k + l = n - 1$. You decide to generate two random keys and combine them with an additional bit. Does the resulting encryption scheme E_{k_1, k_2} , defined as follows, provide perfect secrecy?

$$E_{k_1, k_2}(m) = (k_1 \parallel 1 \parallel k_2) \oplus m,$$

where $m \in \{0, 1\}^n$, $k_1 \in \{0, 1\}^k$, $k_2 \in \{0, 1\}^l$, and $\parallel$ denotes concatenation. What if instead, the message space is modified to only include messages for which the $(k + 1)$ th bit is 0 (i.e., $\mathcal{M} = \{m_1 \parallel 0 \parallel m_2 \mid m_1 \in \{0, 1\}^k, m_2 \in \{0, 1\}^l\}$)?

6. (10 points) Assume we require only that an encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ satisfy the following: For all $m \in \mathcal{M}$, we have $\Pr[\text{Dec}_K(\text{Enc}_K(m)) = m] \geq 2^{-t}$. (This probability is taken over the choice of the key as well as any randomness used during encryption.) Show that perfect secrecy can be achieved with $|\mathcal{K}| < |\mathcal{M}|$ when $t \geq 1$.